

Let the Machine Reveal Itself

A first-principles construction of the tokamak - Sections 1-18

Purpose of this excerpt

This packet presents the opening act of a larger working thesis. It does not begin with a finished tokamak. It starts with controlled D-T fusion and allows each failure of the simplest attempted design to force the next field, component, or parameter into existence.

The governing philosophy

Minimum assumptions, not minimum physics. Every load-bearing statement is derived, identified as a model, or tied to a source. The machine gradually reveals itself rather than being presented as an unexplained object.

What the 28-page excerpt contains

- Fusion temperature, plasma formation, and magnetic steering from the Lorentz force.
- The failure of a straight magnetic corridor and the geometric necessity of a torus.
- Toroidal drift, plasma current, the poloidal field, safety factor, and equilibrium control.
- Heating, elongation, current-profile control, and the first conditional performance scaling.
- A Section 18 ledger stating exactly what has and has not yet been derived.

Reading request

Please assess the causal method, accessibility, and level of rigor. This excerpt is a progress sample, not a claim that the complete reactor or every later engineering constraint has already been closed.

Where the full draft is going

The complete working PDF is substantially longer than this packet. The excerpt was chosen because Section 18 forms a natural first-act conclusion and remains elementary enough for careful verification in one or two days.

1. Derive the machine and its vocabulary

Sections 1-50 derive the fields, geometry, current, shaping, confinement quantities, nuclear island, and magnet radial build.

2. Reconstruct the Freidberg design argument

Sections 51-101 rebuild the 2015 conventional reference, expose its assumptions, test its constraints, and compare its proposed remedies.

3. Test the low-aspect-ratio branch

Sections 102-111 show why lower aspect ratio changes the kink and bootstrap mechanisms. The published candidate closes the kink test; bootstrap closure remains conditional on independently affordable current drive.

4. Confront the centre-stack bargain

Sections 112-127 derive the separate radial, current-density, stress, shielding, startup, and HTS obligations created by the low-A geometry.

Questions for discussion

- Is the derive-before-design structure appropriate for the intended reader?
- Which approximations should receive a fuller derivation, a stronger source, or earlier qualification?
- Would the adjacent six-week centre-stack/startup feasibility study be useful to the group?
- Which recent papers, models, or current group problems should define the next research step?

Academic objective

Complete the thesis and earn one useful joint technical paper, with the hope of sustaining collaboration during a future physics PhD.

Part I — Let the Machine Reveal Itself

A first-principles route to the tokamak

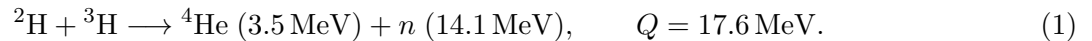
Jony (TUNEM)

Rebuilt opening, August 2026

Promise to the reader. We will not begin by displaying a tokamak and asking you to accept its parts. We will begin with a desired physical event—controlled deuterium–tritium fusion—and let each failure of the simplest attempted design force the next component into existence. Every load-bearing equation will either be derived where it appears or tied to an accessible source. An approximation will be named before it is used.

1. The first requirement: make nuclei approach

The reaction chosen by nearly every near-term fusion-power design is



The released energy is the decrease in rest mass multiplied by c^2 . The products share it in inverse proportion to their masses because momentum is conserved: the lighter neutron carries about four fifths and the alpha particle about one fifth. These numbers will later decide which energy remains in the plasma and which must be caught by a blanket. For now, the obstacle comes before the reward.

A deuteron and triton both carry charge $+e$. At separation r their electrostatic potential energy is

$$U(r) = \frac{e^2}{4\pi\epsilon_0 r}. \quad (2)$$

At $r = 2 \text{ fm}$, a nuclear length scale, Eq. (2) is about 0.72 MeV . A classical particle would need a comparable collision energy to reach that distance. Fusion plasmas operate at much lower thermal energies, typically $T \sim 10\text{--}20 \text{ keV}$, because quantum tunnelling gives some collisions a finite probability of crossing the remaining barrier. Temperature is being measured in energy units: T means $k_B T$. Thus

$$10 \text{ keV} = 10^4 \text{ eV} \quad \Longleftrightarrow \quad T_K = \frac{10^4 \text{ eV}}{k_B} = 1.16 \times 10^8 \text{ K}.$$

This does not mean every ion has the same energy. It means the distribution has a hot tail; fusion occurs disproportionately in that tail. A later section will derive the thermal reaction-rate integral and explain how the energetic tail enters it. Here we need only the first unavoidable conclusion.

Design consequence. Useful D–T fusion requires matter at roughly one hundred million kelvin. Sustained contact with a material surface would cool and contaminate the plasma, while reactor heat flux would erode and damage the surface. The fuel must therefore be kept in vacuum and away from material walls.

Reaction data and the reactor-design context are reproduced in J. P. Freidberg, F. J. Mangiarotti, and J. Minervini, “Designing a Tokamak Fusion Reactor—How Does Plasma Physics Fit In?” (2015), open manuscript: https://library.psfc.mit.edu/catalog/reports/2010/15ja/15ja021/15ja021_full.pdf.

2. The handle nature gives us: charged particles

At 10 keV, the thermal energy is hundreds of times larger than the 13.6 eV binding energy of ground-state hydrogen. Deuterium and tritium do not remain neutral atoms: electrons are stripped from their nuclei. The result is a *plasma*, containing positive ions and negative electrons.

The plasma is not, as a whole, a charged ball. On length scales large compared with a small electrostatic screening length, its positive and negative charge densities nearly cancel: it is *quasineutral*. The useful fact is subtler. Its constituent particles carry charge, so electromagnetic fields can redirect them without a material surface.

For a particle of mass m , charge q , and velocity $\mathbf{v}$, the Lorentz equation is

$$m \frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (3)$$

Set $\mathbf{E} = 0$ temporarily and take a uniform field $\mathbf{B} = B\hat{\mathbf{z}}$. Dotting Eq. (3) with $\mathbf{v}$ gives

$$\frac{d}{dt} \left(\frac{1}{2} m |\mathbf{v}|^2 \right) = q \mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0.$$

The magnetic force changes direction but not kinetic energy. It is therefore a steering force, not a refrigerator and not a source of free plant power.

Write $w = v_x + iv_y$. The perpendicular components of Eq. (3) become

$$\dot{v}_x = \frac{qB}{m} v_y, \quad \dot{v}_y = -\frac{qB}{m} v_x, \quad \implies \quad \dot{w} = -i \frac{qB}{m} w.$$

Hence $w(t) = w(0)e^{-iqBt/m}$: the perpendicular velocity rotates. Integrating once, the particle follows a circle of radius

$$\omega_c = \frac{|q|B}{m}, \quad r_L = \frac{v_\perp}{\omega_c} = \frac{mv_\perp}{|q|B}. \quad (4)$$

Meanwhile the parallel component obeys $\dot{v}_z = 0$. The full orbit is a helix: a small circle across the field combined with free streaming along it.

For a 10 keV deuteron in 3 T, $v_\perp = \sqrt{2T/m_D} = 9.8 \times 10^5$ m/s and $r_L = 6.8$ mm. A magnetic field can therefore keep transverse excursions small compared with a metre-scale chamber. Equation (4), however, exposes the failure: it does nothing to stop motion along the field.

Design consequence. The “invisible wall” cannot be a wall in the ordinary sense. It must be a set of magnetic field lines that guide charged particles, and those lines must eventually have no open ends.

3. First attempted machine: a straight magnetic corridor

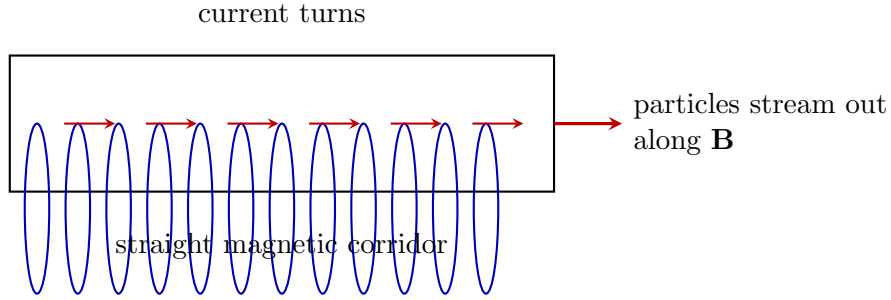
Before closing the ends, we must make a field. Electric current is its source. In a static situation Ampère's law is

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{linked}}. \quad (5)$$

Wind wire into many circular turns around a long cylinder. If there are n turns per unit length, take a rectangular Ampère loop with one long side of length L inside the winding and the other far outside. For an ideal long solenoid, the exterior field is negligible and the two short sides are perpendicular to $\mathbf{B}$. Equation (5) then gives

$$BL = \mu_0(nL)I, \quad \text{so} \quad B \simeq \mu_0 nI. \quad (6)$$

This is our first actual component: a current-carrying coil. Its job is not yet to heat or shape the fuel. Its job is to create the field that makes the gyro-orbits of Eq. (4).



The corridor controls cross-field motion, but Eq. (3) already told us that $v_{\parallel}$ remains free. At the deuteron speed above, a particle crosses a 10 m device in roughly

$$t_{\text{end}} \sim \frac{L}{v_{\parallel}} \sim \frac{10 \text{ m}}{10^6 \text{ m/s}} = 10 \mu\text{s}.$$

Making the straight corridor stronger reduces r_L but does not change this end-loss time. Magnetic mirrors can reflect some pitch angles, but particles within their loss cone still escape. We have learned something more valuable than a failed design: the problem is topological, not merely quantitative.

Design consequence. Do not try to plug the ends of the magnetic corridor. Remove the ends by joining them. The simplest closed version of a tube is a torus.

Model and limits. Equation (6) assumes a solenoid long compared with its radius. End fields and finite-coil corrections are deliberately omitted because the calculation is being used to expose, not conceal, end loss.

4. Closing the corridor creates the torus

Bend the straight magnetic corridor until its two ends meet. The field now runs the long way around a torus; this direction is called *toroidal*. A cross-section of the tube has radius a , while the distance from the symmetry axis to the centre of that cross-section is the major radius R_0 :

$$A \equiv \frac{R_0}{a} \quad (\text{aspect ratio}). \quad (7)$$

These parameters have appeared because the design has acquired two geometrically different lengths, not because a tokamak diagram handed them to us.

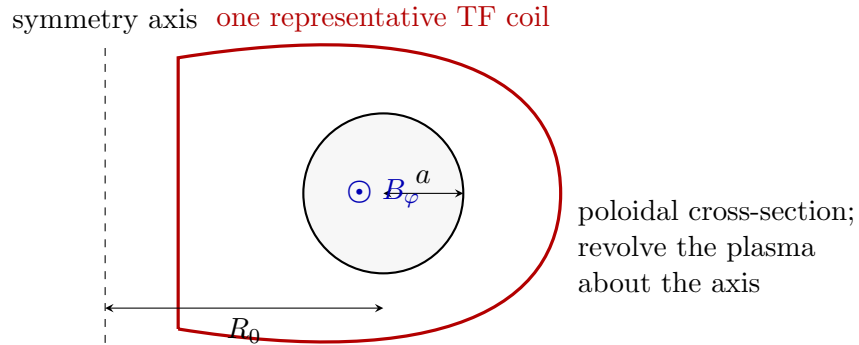
Primary Friedberg parameter — a (minor radius) and R_0 (major radius). Closing the straight corridor makes these two dimensions unavoidable: a measures the cross-sectional reach available to the plasma, while R_0 measures the radius of the closed path that removes end loss. Their ratio $A = R_0/a$ is useful, but is a derived supporting quantity rather than an additional primary parameter.

How do coils make a toroidal field? Arrange current-carrying loops around the torus so that, in the ideal limit of continuously distributed coils, the system is axisymmetric. Apply Ampère's law to a circle of radius R centred on the symmetry axis. The field is tangent to the circle and constant along it, while the loop links total current NI_c :

$$2\pi R B_\varphi(R) = \mu_0 N I_c, \quad \boxed{B_\varphi(R) = \frac{\mu_0 N I_c}{2\pi R}}. \quad (8)$$

Thus the toroidal field is intrinsically stronger on the inboard side and weaker on the outboard side. Across a circular plasma,

$$\frac{B_{\text{in}}}{B_{\text{out}}} = \frac{R_0 + a}{R_0 - a} = \frac{A + 1}{A - 1}. \quad (9)$$



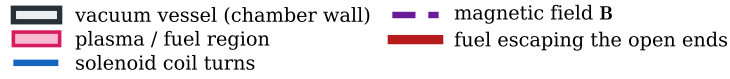
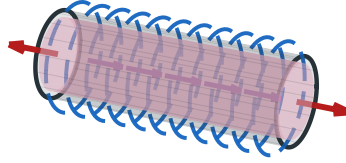
We have now removed the ends. We have not yet built a tokamak. The object presently on the page is only a toroidal magnetic bottle, and Eq. (8) has introduced a new asymmetry: the field strength varies across the tube. The next question is whether this closed field actually confines a particle.

Model and limits. Equation (8) is exact for the axisymmetric continuous-coil idealization. A finite number of real toroidal-field coils produces ripple and a small external stray field; those are later engineering corrections, not erased facts.

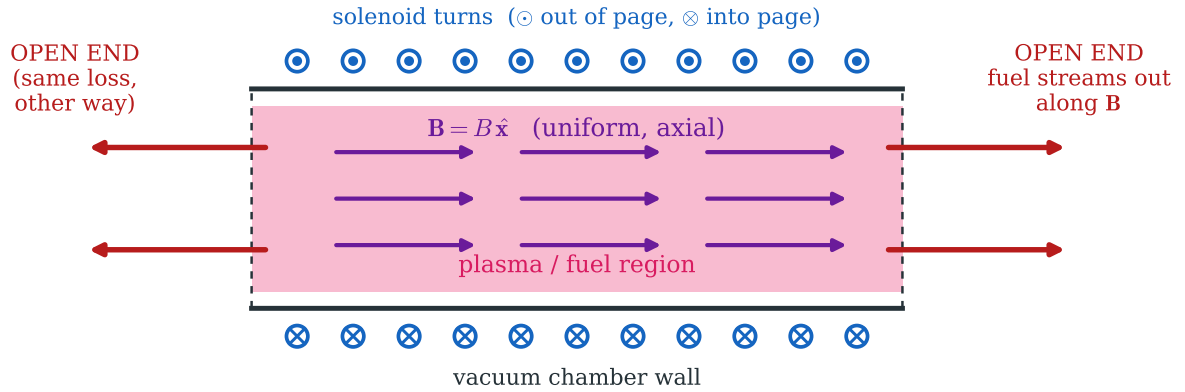
Stage 1 — the straight magnetic corridor

a solenoid makes $B_{\parallel}$; the ends are open, so the fuel leaves

perspective view



longitudinal section



Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 1: The first machine state. A solenoid makes the field; the corridor has open ends, which is the failure the next section names.

5. The closed bottle still leaks: toroidal drifts

The gyro-orbit derivation assumed uniform $\mathbf{B}$. Equation (8) makes the field stronger on one side of every orbit than the other, and the field line itself is curved. We now calculate the leading effect, assuming $r_L/R_0 \ll 1$ so the field changes little during one gyro-orbit.

First derive a useful general result. Add a constant force $\mathbf{F} \perp \mathbf{B}$ to Eq. (3), with $\mathbf{E} = 0$. Seek a solution consisting of gyromotion plus a constant velocity $\mathbf{v}_F$. Requiring the constant terms to cancel gives

$$q\mathbf{v}_F \times \mathbf{B} + \mathbf{F} = 0 \quad \Longrightarrow \quad \boxed{\mathbf{v}_F = \frac{\mathbf{F} \times \mathbf{B}}{qB^2}}. \quad (10)$$

The torus supplies two effective forces. Conservation of the magnetic moment $\mu = mv_\perp^2/(2B)$ is not merely asserted: a gyro-orbit is a current loop, so

$$\mu = \left(\frac{q\omega_c}{2\pi}\right) (\pi r_L^2) = \frac{mv_\perp^2}{2B}.$$

When B changes little during one orbit, gyro-averaging makes μ an adiabatic invariant; the orbit therefore has potential energy μB and gradient force $\mathbf{F}_{\nabla B} = -\mu \nabla B$.¹ Following a field line curved with radius R also requires centripetal acceleration; in the frame of the guiding centre this enters as the outward curvature force $\mathbf{F}_c = (mv_\parallel^2/R)\hat{\mathbf{R}}$.

For Eq. (8), $\nabla B = -(B/R)\hat{\mathbf{R}}$. Therefore both effective forces point radially outward. With $\mathbf{B} = B\hat{\boldsymbol{\varphi}}$ and $\hat{\mathbf{R}} \times \hat{\boldsymbol{\varphi}} = \hat{\mathbf{Z}}$, Eq. (10) gives

$$\begin{aligned} \mathbf{v}_{\nabla B} &= \frac{mv_\perp^2}{2qBR} \hat{\mathbf{Z}}, \\ \mathbf{v}_c &= \frac{mv_\parallel^2}{qBR} \hat{\mathbf{Z}}, \\ \therefore \quad \boxed{\mathbf{v}_d &= \frac{m}{qBR} \left(v_\parallel^2 + \frac{1}{2} v_\perp^2 \right) \hat{\mathbf{Z}}}. \end{aligned} \quad (11)$$

Equation (11) is fatal to the design in its present form. Positive and negative charges drift in opposite vertical directions because the sign is set by q . The resulting charge separation creates a vertical electric field. Returning to Eq. (3), an electric force $q\mathbf{E}$ inserted into Eq. (10) gives

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}, \quad (12)$$

which is independent of charge and mass. Ions and electrons now move together. For the vertical charge-separation field produced here, that common drift is radially outward, toward the vessel wall.

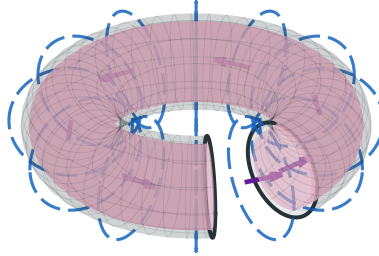
Design consequence. Closing the straight field into a torus removes end loss but creates systematic toroidal drifts. A purely toroidal magnetic field cannot confine the plasma. The field line must be made to visit both the top and bottom of the torus so that the same vertical drift alternates between outward and inward displacement.

¹A systematic open derivation of guiding-centre motion and its ordering assumptions is A. J. Brizard and T. S. Hahm, “Foundations of nonlinear gyrokinetic theory,” <https://arxiv.org/abs/physics/0409119>.

Stage 2 — the corridor is closed into a torus

the same tube bent end-to-end; TF coils now make $B_\varphi \propto 1/R$

perspective view (vessel cut open)

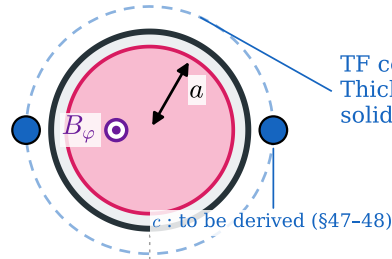


vacuum vessel (chamber wall)
 toroidal-field (TF) coils

plasma / fuel region
 magnetic field B

symmetry axis

poloidal cross-section (R - Z plane)



TF coil seen edge-on (it pierces this plane twice). Thickness c : TO BE DERIVED (§47-48). The coil is solid because §4 earned it; only c is still open.

R_0 major radius

$\odot$: the vector points out of the page

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 2: The corridor closed into a torus. The straight solenoid turns have become toroidal-field coils around the same tube; R_0 , a and B_φ now exist because the geometry acquired two lengths.

6. A second field is forced upon us

A field line confined to one horizontal plane experiences the drift of Eq. (11) with the same radial consequence throughout its journey. Make the line wind the short way around the tube as well as the long way around the torus. It then passes successively through the top and bottom of the cross-section; a vertical displacement is outward on one half and inward on the other. The unbounded drift is replaced, to leading order, by a bounded orbit excursion.

The required short-way field is called *poloidal*. There are two broad ways to make it. External helical windings can do so, producing a stellarator. The tokamak makes the main poloidal field with a toroidal electric current carried by the plasma itself. On a circular cross-section, apply Ampère's law to a poloidal circle of radius r enclosing plasma current $I_{\text{enc}}(r)$:

$$2\pi r B_\theta(r) = \mu_0 I_{\text{enc}}(r), \quad \boxed{B_\theta(r) = \frac{\mu_0 I_{\text{enc}}(r)}{2\pi r}}. \quad (13)$$

The combined field

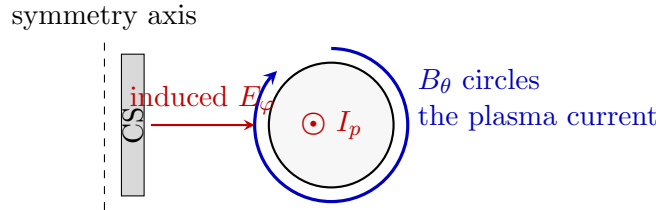
$$\mathbf{B} = B_\varphi \hat{\boldsymbol{\varphi}} + B_\theta \hat{\boldsymbol{\theta}}$$

is helical. A field line no longer remains at one height. The toroidal drift problem has dictated a plasma current, and the plasma current has dictated a helical field.

Where does that current come from? Place a solenoid through the central hole and vary its magnetic flux. Faraday's law,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}, \quad (14)$$

creates a toroidal electric field. The plasma acts as the one-turn secondary winding of a transformer; the induced electric field drives I_p . The central solenoid has therefore not been added because tokamaks traditionally possess one. It has been demanded by the decision to obtain the poloidal field through plasma current.



Design consequence. The machine revealed so far contains a vacuum vessel, toroidal-field coils, a toroidal plasma, and a means of driving plasma current. This is the minimal physical chain that creates the tokamak field. It is not yet a reactor: we have not shown pressure equilibrium, stability, adequate confinement time, heating, exhaust, breeding, magnet survival, or net electricity. Each of those failures will demand its own component.

One distinction will matter later. The plasma current produces the principal poloidal field that makes field lines helical. Separate external poloidal-field coils will be introduced only when radial force balance, position control, and shaping require them. The next derived parameter will be the pitch of the helix—the safety factor q —because the current that repairs drift also creates the machine's first major stability limit.

7. The helix has a pitch: the safety factor q

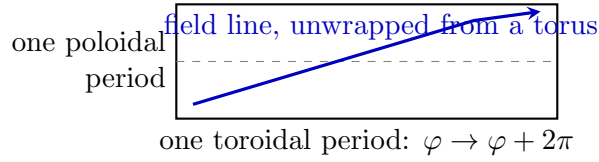
The combined field has two components, so a field line advances two ways at once. On a nearly circular magnetic surface of minor radius r , a short displacement has toroidal length $R d\varphi$ and poloidal length $r d\theta$. Because the displacement is parallel to $\mathbf{B}$, their ratio must equal the corresponding ratio of field components:

$$\frac{R d\varphi}{r d\theta} = \frac{B_\varphi}{B_\theta}. \quad (15)$$

At large aspect ratio the variation of R and B_φ around one cross-section is small enough that $R \simeq R_0$ and $B_\varphi \simeq B_0$. One poloidal circuit, $\Delta\theta = 2\pi$, then advances the field line toroidally by $2\pi q$, where

$$q(r) \equiv \frac{\Delta\varphi}{2\pi} \simeq \frac{r B_0}{R_0 B_\theta(r)}. \quad (16)$$

Thus q is simply the number of long-way circuits made per one short-way circuit. The word *safety* is historical; the stability reason for that name comes later.



Substitute the poloidal field already derived in Eq. (13):

$$q(r) \simeq \frac{2\pi r^2 B_0}{\mu_0 R_0 I_{\text{enc}}(r)}, \quad q_a \simeq \frac{2\pi a^2 B_0}{\mu_0 R_0 I_p}. \quad (17)$$

The edge value q_a has not been chosen by a convention. It has emerged from geometry, Ampère's law, and the decision to use plasma current for the poloidal field. Reading the same relation backwards gives the equally important current relation

$$I_p \simeq \frac{2\pi a^2 B_0}{\mu_0 R_0 q_a}. \quad (18)$$

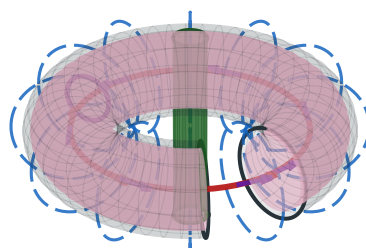
Any later lower bound on q_a is therefore an upper bound on plasma current. This connection is one of the load-bearing constraints in the later reactor audit.

A rational value $q = m/n$, for integers m, n , closes after n poloidal and m toroidal turns. An irrational value never exactly closes and instead samples its surface.

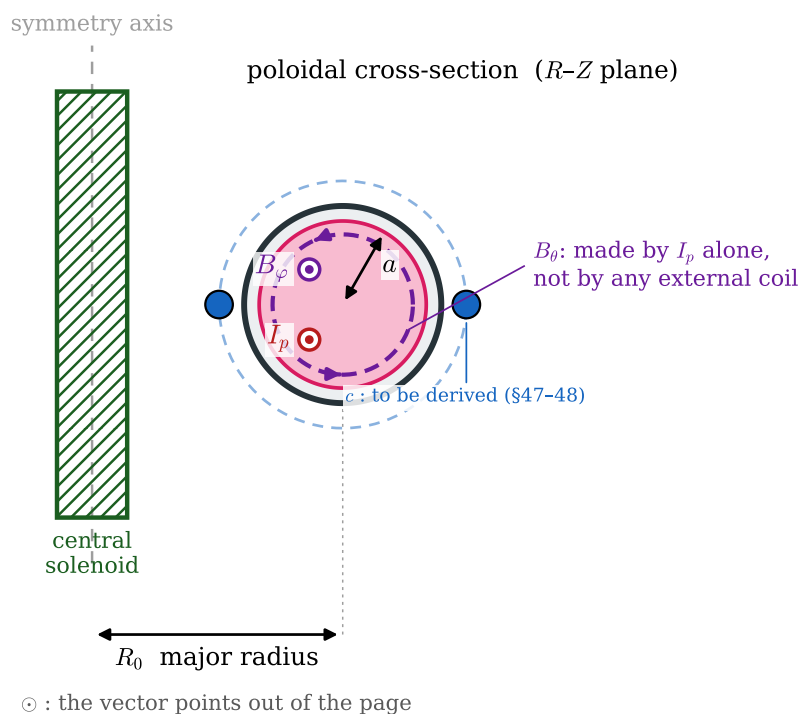
Design consequence. The toroidal field, the plasma current, the size of the torus, and the helix pitch are not independent knobs. Equation (18) ties them together. The current that makes confinement possible has already spent part of the machine's future stability budget.

The shaped generalization and later stability use are given by Freidberg, Mangiarotti, and Minervini (2015), Eq. (30): https://library.psfc.mit.edu/catalog/reports/2010/15ja/15ja021/15ja021_full.pdf.

perspective view (vessel cut open)



- | | | | |
|---|------------------------------|---|-----------------------|
| | vacuum vessel (chamber wall) | | central solenoid (CS) |
|  | plasma / fuel region |  | magnetic field B |
|  | toroidal-field (TF) coils |  | plasma current I_p |



Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak build.py.

Figure 3: The tokamak field. A central solenoid drives I_p , and I_p alone supplies B_θ — no external coil does this. Every component of the previous state is retained.

8. A hot plasma is not a passenger: it has pressure

Until now we have followed one particle in fields prescribed from outside. A fusion plasma contains enough particles that its collective push matters. For one ideal gas species, random impacts transfer momentum to a surface and give $p = nk_B T$. We are using energy units for temperature, so this is simply $p = nT$. Electrons and ions each contribute; quasineutrality gives $n_e \simeq n_i \equiv n$, hence

$$p = n_e T_e + n_i T_i \simeq 2nT \quad \text{when } T_e \simeq T_i \equiv T. \quad (19)$$

For orientation, $n = 10^{20} \text{ m}^{-3}$ and $T = 10 \text{ keV}$ give $p \simeq 3.2 \times 10^5 \text{ Pa}$, about three atmospheres. This pressure is inside a near-vacuum vessel, so it tries to expand the plasma.

At scales large compared with a gyro-orbit, it is useful to average the particles into one conducting fluid. Its momentum equation is

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p + \mathbf{J} \times \mathbf{B}. \quad (20)$$

Here $-\nabla p$ is the force due to pressure and $\mathbf{J} \times \mathbf{B}$ is the magnetic force density. A static equilibrium requires

$$\boxed{\mathbf{J} \times \mathbf{B} = \nabla p.} \quad (21)$$

This is a model declaration, not a universal law: it assumes slow evolution, negligible bulk flow, small gyro-orbits compared with the plasma size, and an approximately scalar pressure. Those assumptions will be checked rather than silently inherited when we use the equation quantitatively.

One consequence follows without solving anything. Dot Eq. (21) with $\mathbf{B}$:

$$\mathbf{B} \cdot \nabla p = \mathbf{B} \cdot (\mathbf{J} \times \mathbf{B}) = 0. \quad (22)$$

Pressure is constant along a field line. The invisible wall is therefore not a loose collection of lines: a static confined plasma requires nested magnetic surfaces on which pressure can live.

There is a natural way to compare plasma push with magnetic capability. Build the field of a solenoid slowly. With N turns, length L , area A , and flux linkage NBA , the source supplies $dW = I d(NBA)$. Since $B = \mu_0 NI/L$, integration gives

$$W = \frac{B^2}{2\mu_0} AL. \quad (23)$$

Thus $B^2/(2\mu_0)$ is the magnetic energy per unit volume and has the same units as pressure. Define

$$\boxed{\beta \equiv \frac{p}{B^2/(2\mu_0)}.} \quad (24)$$

It is the fraction of magnetic pressure-scale that the plasma is asking the field to hold. Raising B makes a given pressure easier to support; raising p makes the same magnet work harder.

Design consequence. A tokamak cannot be designed by field-line geometry alone. Its magnetic surfaces must also balance pressure. This forces a second task on the coil system: hold the plasma at a chosen radial position while its own pressure and current try to move it.

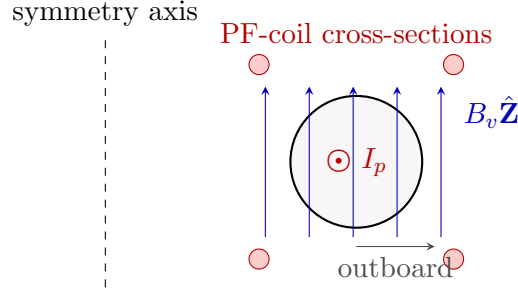
The static-MHD model and the use of β in reactor design are treated in the open Freidberg–Mangiarotti–Minervini manuscript cited on page 1. The field energy relation above is derived from the work needed to establish a solenoidal field.

9. Holding position forces external poloidal-field coils

The plasma current is a toroidal current loop. A current loop in a vertical magnetic field feels a radial force. This is not a slogan; apply the elementary force law $d\mathbf{F} = I d\boldsymbol{\ell} \times \mathbf{B}$ to a circular loop. For $d\boldsymbol{\ell} = R_0 d\varphi \hat{\boldsymbol{\varphi}}$ and $\mathbf{B}_v = B_v \hat{\mathbf{Z}}$,

$$d\mathbf{F} = I_p R_0 B_v d\varphi \hat{\mathbf{R}}, \quad \mathbf{F}_R = 2\pi R_0 I_p B_v \hat{\mathbf{R}}. \quad (25)$$

Changing the sign of B_v changes the force from outward to inward.



The plasma has several sources of radial imbalance: its pressure is higher in the interior; its own poloidal field has a curved, toroidal geometry; and the applied toroidal field varies as $1/R$. Equation (21) says that the sum must cancel, but it does *not* say that a free current ring automatically sits at the desired R_0 . Equation (25) gives the actuator that can correct the residual force.

This finally distinguishes two uses of a poloidal field. The field generated by I_p is the principal field that makes field lines helical and repairs drift. The external vertical component shown here is part of an applied *poloidal-field* (PF) system. Its first job is equilibrium and radial position. The same system will later acquire separate coils for shaping and fast feedback.

We deliberately do not yet quote one “required vertical field” formula. Its exact coefficient depends on the pressure profile, current profile, cross-section, and the plasma’s self-inductance. Giving a single number before those objects have been derived would conceal those assumptions. At this stage physics has rigorously forced the existence and the direction of the control field; the full equilibrium calculation will set its magnitude.

Design consequence. The first coils around the torus made B_φ . The central solenoid drove I_p . Pressure balance now forces a separate external PF-coil system: it must hold the plasma’s position rather than merely create field lines.

10. Flux surfaces turn coil currents into a plasma boundary

To say that coils “hold position” is still too vague. Axisymmetry lets us name the geometric object they control. In cylindrical coordinates (R, φ, Z) , with no φ dependence, $\nabla \cdot \mathbf{B} = 0$ becomes

$$\frac{\partial(RB_R)}{\partial R} + \frac{\partial(RB_Z)}{\partial Z} = 0. \quad (26)$$

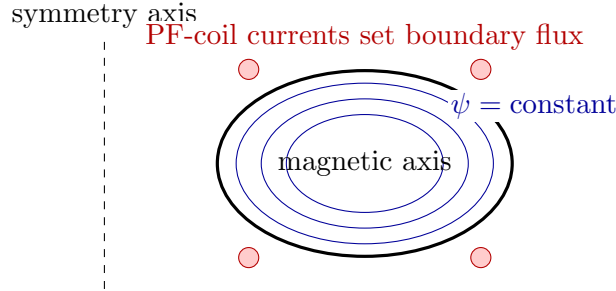
This is the two-dimensional statement that a flow has no sources or sinks. Therefore there is a scalar function $\psi(R, Z)$ whose derivatives generate the poloidal field:

$$B_R = -\frac{1}{R} \frac{\partial \psi}{\partial Z}, \quad B_Z = \frac{1}{R} \frac{\partial \psi}{\partial R}, \quad \mathbf{B}_p = \frac{\nabla \psi \times \hat{\varphi}}{R}. \quad (27)$$

Substituting these two derivatives into Eq. (26) verifies the claim in one line. More importantly,

$$\mathbf{B} \cdot \nabla \psi = 0. \quad (28)$$

The toroidal part is also perpendicular to $\nabla \psi$ because ψ does not depend on φ . Thus a field line stays on a surface $\psi = \text{constant}$. The quantity $2\pi\psi$ is the poloidal magnetic flux enclosed by the toroidal loop at (R, Z) .



The outer closed contour is not just the vessel wall. It is the last closed magnetic surface selected by the plasma current together with the PF-coil currents. The vessel must contain and protect this arrangement, but it cannot replace the magnetic boundary: a conducting wall cannot hold a 10 keV plasma in steady contact.

This is also why shape will matter. Moving PF-coil currents changes the boundary value of ψ , and therefore changes the cross-section of the nested surfaces. A circle is only the first solvable shape, not a physical law. Elongation and triangularity will be introduced only when a specific performance or stability problem demands them.

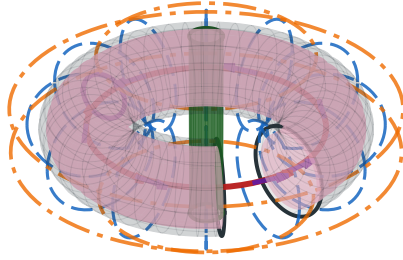
Design consequence. PF coils control flux, and flux contours are the plasma boundary. The emerging machine therefore needs a vacuum vessel outside a deliberately shaped, magnetically defined plasma—not a metal container that touches the fuel.

This axisymmetric flux representation is derived from $\nabla \cdot \mathbf{B} = 0$ above. For an openly available full equilibrium treatment, see Cerfon and Freidberg, “One size fits all” analytic solutions to the Grad–Shafranov equation (2010), [open public manuscript](#).

Stage 4 — equilibrium and position control

external PF coils apply a vertical field B_v to hold the current ring at R_0

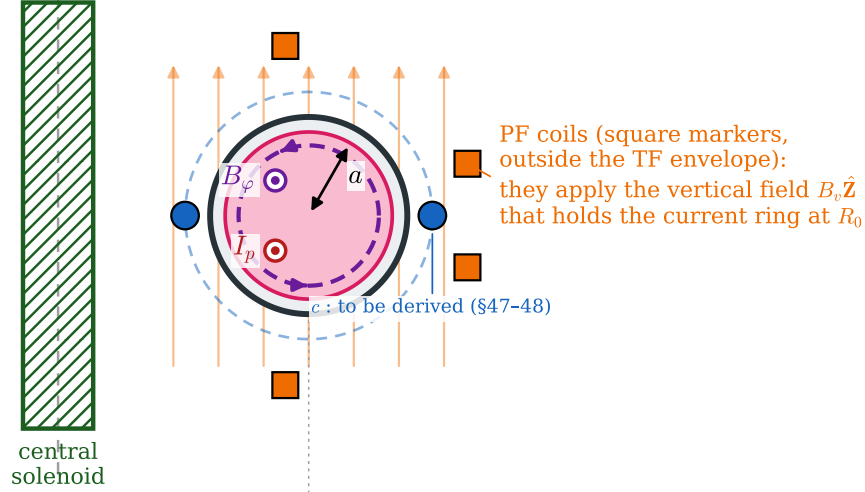
perspective view (vessel cut open)



- | | | | |
|--|------------------------------|--|---------------------------|
| | vacuum vessel (chamber wall) | | poloidal-field (PF) coils |
| | plasma / fuel region | | magnetic field B |
| | toroidal-field (TF) coils | | plasma current I_p |
| | central solenoid (CS) | | |

symmetry axis

poloidal cross-section (R - Z plane)



R_0 major radius

$\odot$: the vector points out of the page

Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 4: Equilibrium and position control. External PF coils, drawn as square markers strictly outside the toroidal-field envelope, apply the vertical field that holds the current ring at R_0 .

Eleven. The transformer has a clock

The central solenoid was introduced to drive the toroidal current. Faraday’s law now reveals a price. Around a toroidal loop,

$$V_{\text{loop}} \equiv \oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_{\text{link}}}{dt}. \quad (29)$$

Integrating through a discharge gives

$$\int V_{\text{loop}} dt = -\Delta\Phi_{\text{link}}. \quad (30)$$

The available linked-flux change has units of volt-seconds. A central solenoid can only swing through a finite magnetic flux before its current, field, and mechanical limits are reached. It can start and sustain a pulse, but it cannot supply a steady DC loop voltage forever. A volt-second is exactly a weber: $1 \text{ Vs} = 1 \text{ Wb}$. It is therefore a unit of linked magnetic flux—the transformer’s finite magnetic-flux budget—not a unit of stored energy.

The induced current initially supplies useful heat. The local electrical power density is $\mathbf{J} \cdot \mathbf{E}$; integrating over the plasma gives

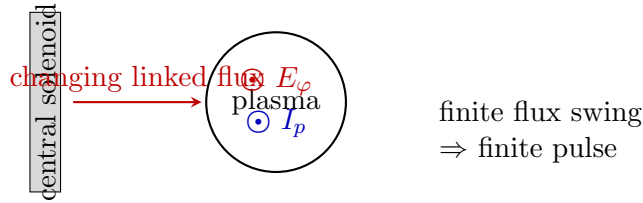
$$P_{\text{ohm}} = \int \mathbf{J} \cdot \mathbf{E} dV \simeq I_p V_{\text{loop}} = I_p^2 R_p. \quad (31)$$

The final form is the lumped-circuit approximation, with plasma resistance R_p . It is useful for the design argument, not a claim that the current density is uniform.

There is a second limit. In a fully ionized, collisional plasma the classical Spitzer resistivity falls approximately as

$$\eta \equiv \frac{E_{\parallel}}{J_{\parallel}} \propto T_e^{-3/2}. \quad (32)$$

Hot electrons collide less effectively, so at comparable current density the plasma becomes a better conductor and ohmic heating weakens. This is a pleasant fact for current flow and an inconvenient fact for trying to heat a reactor plasma by resistance.



Design consequence. The central solenoid has two jobs—start the plasma current and provide early ohmic heat—but neither is an indefinitely scalable heat source. The machine must acquire auxiliary heating, and eventually non-inductive current drive, if it is to operate beyond a transformer pulse.

The $T_e^{-3/2}$ Spitzer scaling and the power-density form ηJ^2 are available openly in the NRL Plasma Formulary: https://www.nrl.navy.mil/Portals/38/PDF%20Files/NRL_Plasma_Formulary_2019.pdf.

12. Reaching fusion temperature forces a heating system

The statement “make the fuel hot” on page 1 can now be made into an energy balance. Each Maxwellian species has thermal energy density $3n_s T_s/2$. For an approximately uniform D–T plasma with equal ion and electron temperatures,

$$W \simeq 3nTV, \quad P_{\text{loss}} \equiv \frac{W}{\tau_E}. \quad (33)$$

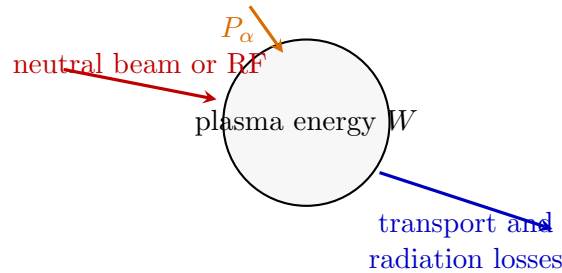
The first relation is energy stored in the plasma. The second *defines* the energy-confinement time τ_E : if all heating were removed, it is the time scale on which the stored energy would be lost at the present loss rate.

The bookkeeping law is therefore

$$\frac{dW}{dt} = P_{\text{aux}} + P_{\alpha} - P_{\text{loss}} - P_{\text{rad}}. \quad (34)$$

At startup P_{α} is small because fusion is small. Auxiliary power must raise W faster than transport and radiation remove it. Later, the 3.5 MeV alpha particles from Eq. (1) are charged and can be magnetically confined; their deposited energy is the desired self-heating term P_{α} . The 14.1 MeV neutrons are neutral and escape, eventually to be captured in a blanket.

Why not simply fire charged particles into the plasma? The same magnetic field that confines the fuel bends an incoming charged beam before it reaches the core. A neutral beam solves that entry problem: neutral atoms cross the field, ionize in the plasma, and then transfer their energy and momentum by collisions. Radio-frequency systems use another handle already present in Eq. (3): an oscillating electric field can transfer energy resonantly to selected charged-particle orbits. These are not ornaments on a completed tokamak. They are responses to the transformer and power-balance limits just derived.



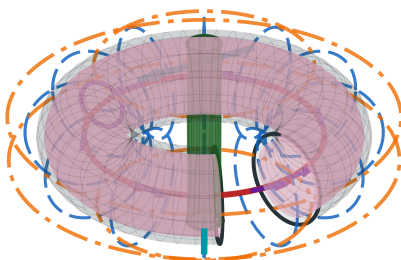
The next question is not yet whether the machine makes net electricity. It is more basic: how much pressure can a magnetic equilibrium and a stable plasma sustain, and how should the boundary be shaped to improve that answer? Those questions will force elongation, triangularity, feedback, and the first genuine stability limits. Only after the ordinary tokamak has been derived on those terms will it make sense to ask whether a spherical tokamak changes the bargain.

Design consequence. Twelve pages in, the machine has emerged component by component: a vacuum environment, toroidal-field coils, a plasma current and central solenoid, external PF coils, and auxiliary heating. What remains absent is just as important: the shaped boundary, exhaust, blanket, shielding, superconducting magnets, and the reactor’s energy ledger must each earn their place through the next physical failure.

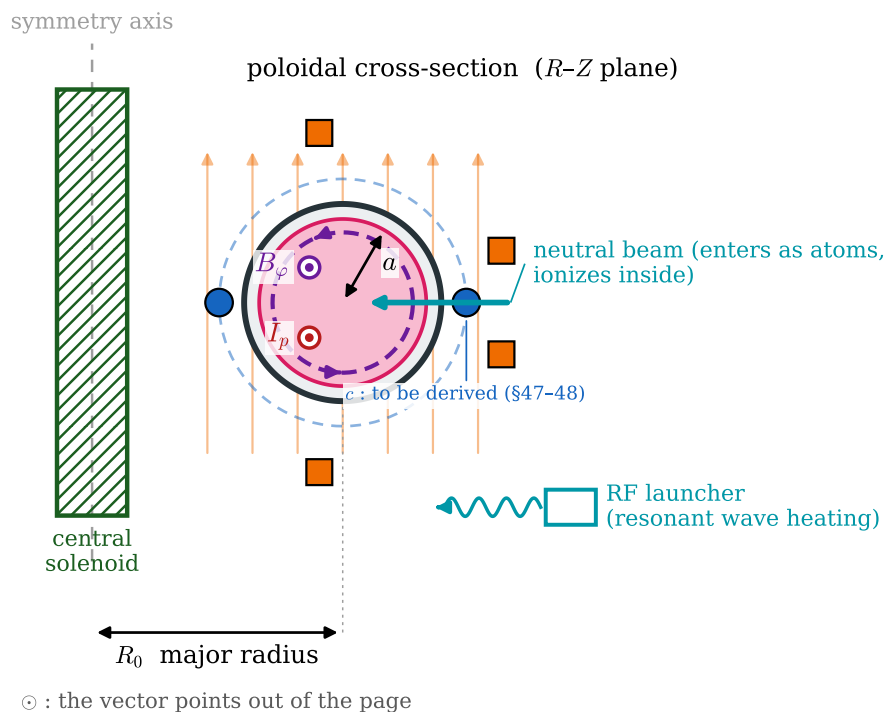
Stage 5 — auxiliary heating

a neutral beam and an RF launcher add power the transformer cannot

perspective view (vessel cut open)



- | | | | |
|---|------------------------------|---|---------------------------|
|  | vacuum vessel (chamber wall) |  | poloidal-field (PF) coils |
|  | plasma / fuel region |  | magnetic field B |
|  | toroidal-field (TF) coils |  | plasma current I_p |
|  | central solenoid (CS) |  | auxiliary heating ports |



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.00$, $\delta = 0.00$, 12 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 5: Auxiliary heating. A tangential neutral beam and a radial RF launcher add the power the transformer cannot. No blanket, divertor or shield appears: none has been derived.

13. More room is useful; a taller cross-section supplies it

The circular cross-section used so far was deliberately austere. It makes the geometry easy to see, but it is not the only allowed shape. Suppose the plasma boundary is an ellipse with horizontal semi-axis a and vertical semi-axis κa . The dimensionless number

$$\kappa \equiv \frac{\text{vertical semi-axis}}{\text{horizontal semi-axis}} \quad (35)$$

is called *elongation*. A circle has $\kappa = 1$; a vertically stretched plasma has $\kappa > 1$. Parameterize the boundary by

$$R(\theta) = R_0 + a \cos \theta, \quad Z(\theta) = \kappa a \sin \theta.$$

The cross-sectional area is $\pi(\kappa a)a = \pi\kappa a^2$. Rotating that area once around the major radius R_0 gives, to leading order in a/R_0 ,

$$V = 2\pi R_0 (\pi\kappa a^2) = 2\pi^2 R_0 a^2 \kappa. \quad (36)$$

Thus, at fixed R_0 and a , elongation buys volume linearly. More volume can hold more fuel and more fusion power; later we will state exactly which other quantities must be held fixed before making that comparison.

Primary Friedberg parameter — κ (elongation). Equation (35) changes the plasma volume in Eq. (36) and enters the shaped safety-factor and current limits derived below. It is not decorative geometry: it is a design lever with a stability cost.

The price is immediate. A round set of poloidal-field coils naturally supports a round boundary. An elliptical boundary needs an intentionally nonuniform external field: some coils pull or push more strongly above and below the midplane than at the outboard side. The PF-coil system derived on [Sec. 9](#) therefore acquires a second job. It does not merely keep the plasma centered; it helps choose the boundary shape. This is the first time an earlier component has been asked to do more than solve the failure that first called it into existence.

FIGURE GUIDE

The cumulative elongation diagram appears on the following page.

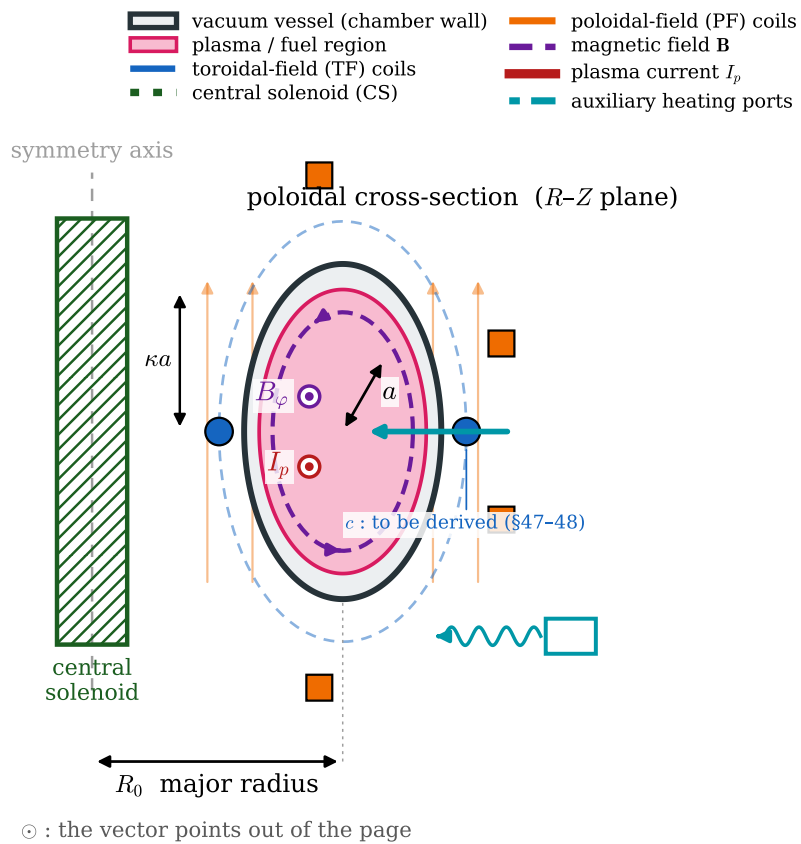
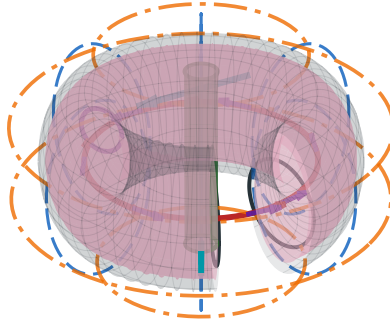
Model and limits. Equation (36) treats the cross-section as an ellipse and the torus as thin enough that its circumference is $2\pi R_0$. Real equilibria also use triangularity and have shifted magnetic surfaces. Those refinements come only after the basic geometric benefit and cost of κ are clear.

Design consequence. If a reactor needs more plasma volume without simply increasing every linear dimension, it has a reason to elongate the cross-section. That choice requires shaping PF coils and raises a new question: is the taller plasma passively stable?

Stage 6 — the boundary is elongated ($\kappa > 1$)

a taller cross-section buys area, hence I_p at fixed q_* ; the coil set follows the same boundary

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.

$R_0/a=3.0$, $\kappa=1.70$, $\delta=0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 6: Machine state: elongation. A taller cross-section supplies volume without growing every linear dimension, and the coil set follows the same boundary. Every component of the previous state is retained.

14. A taller plasma can tip: shaping requires feedback

Imagine displacing an elongated plasma a small distance z upward. The magnetic forces from the surrounding coils change. Near a chosen equilibrium height, any smooth vertical force can be expanded as

$$F_Z(z) = F_Z(0) + \left. \frac{\partial F_Z}{\partial z} \right|_0 z + \dots \quad (37)$$

At equilibrium $F_Z(0) = 0$. Newton's law then becomes

$$M\ddot{z} = K_z z, \quad K_z \equiv \left. \frac{\partial F_Z}{\partial z} \right|_0. \quad (38)$$

If $K_z < 0$, Eq. (38) is an oscillator: a displacement creates a restoring force. If $K_z > 0$, its solutions include $z \propto e^{\sqrt{K_z/M}t}$: a small upward displacement grows instead of returning. This elementary sign test is the core of a vertical instability. Elongation generally makes this control problem harder, because a tall current-carrying column has a stronger tendency to move as a whole.

This does *not* make elongation forbidden. It says that its equilibrium cannot be trusted to remain in place by itself. A real machine measures its position with magnetic sensors, compares it with a requested position, and changes currents in selected PF coils. The control system must act before the growing displacement reaches the wall. The causal chain is therefore precise: more useful shape $\rightarrow$ weaker passive vertical stability $\rightarrow$ sensors, power supplies, and fast controlled coils.

FIGURE GUIDE

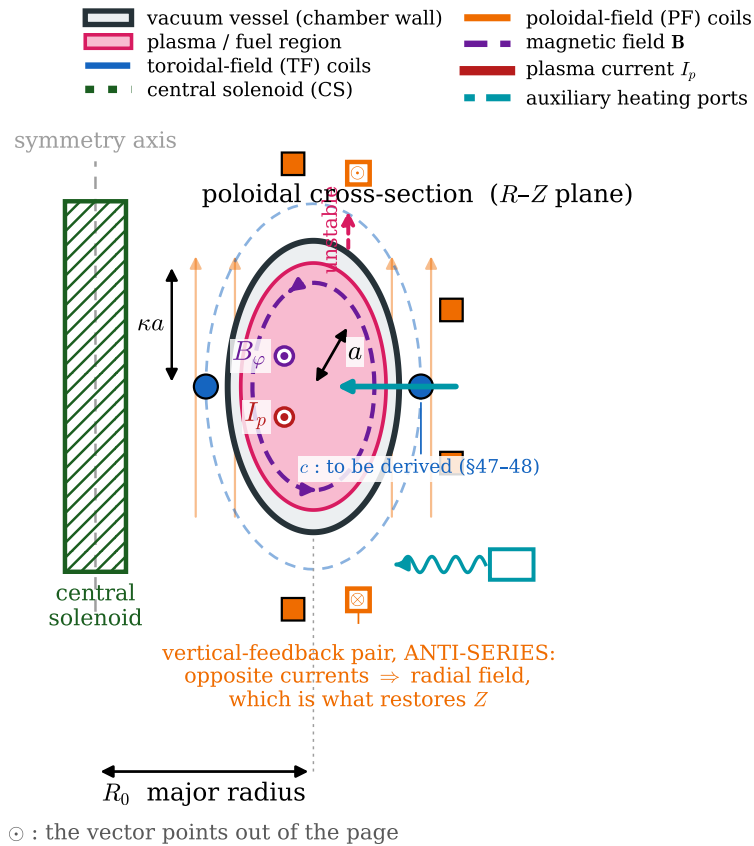
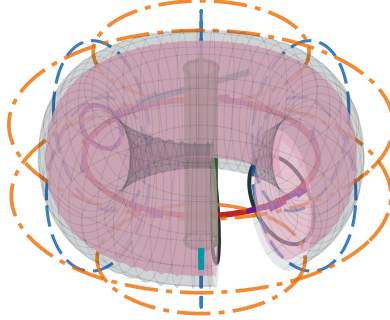
The cumulative feedback-control diagram appears on the following page.

Model and limits. Equation (38) is a one-coordinate linear model. Actual vertical control couples plasma shape, conducting structures, coil circuits, and sensor delays. It is used here only to establish the sign logic that makes feedback necessary. For an openly available treatment of elongation and vertical stability, see E. Cerfon, J. P. Freidberg, and J. P. Vesey, “Tokamak elongation: how much is too much? Part 1,” <https://www.cambridge.org/core/services/aop-cambridge-core/content/view/1ABC7975053E19C0A93D9952CBB53102/S0022377815001270a.pdf>.

Design consequence. A shaped tokamak is not merely coils plus plasma. It includes a closed feedback system. The allowed elongation is set jointly by the gain in volume, the available PF-coil authority, nearby conducting structures, and the speed and reliability of active control.

Stage 7 — elongation is unstable, so it must be fed back
an anti-series coil pair supplies the radial field that opposes vertical displacement

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 7: Machine state: vertical control. The pair is wired *anti-series* ($\odot$ above, $\otimes$ below): opposite currents make the combined field radial on the midplane, which is what restores Z . A same-sign pair would only move the plasma in R .

15. The same geometry changes how much plasma current is allowed

We already derived a safety factor for a simple circular plasma. It counted, in effect, how far a field line moves toroidally while making one poloidal circuit. The local version at minor radius r is

$$q(r) = \frac{2\pi r^2 B_0}{\mu_0 R_0 I_{\text{enc}}(r)}, \quad I_{\text{enc}}(r) = 2\pi \int_0^r j_\varphi(r') r' dr'. \quad (39)$$

Here j_φ is the toroidal current density and I_{enc} is the current inside the selected magnetic surface. The first relation follows from the same field-line ratio used before, $q \simeq r B_\varphi / (R_0 B_\theta)$; the second is the area integral of current density, and Ampère's law gives $B_\theta \simeq \mu_0 I_{\text{enc}} / (2\pi r)$.

An elongated boundary alters this simple geometry. A widely used large-aspect-ratio, elongated-plasma estimate is

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I_p} \frac{1 + \kappa^2}{2}. \quad (40)$$

The first fraction is the circular edge estimate. The factor $(1 + \kappa^2)/2$ records that one poloidal circuit is no longer circular. Equation (40) is not being introduced as a mysterious correction: it is the next-order geometry that becomes necessary once we deliberately choose $\kappa \neq 1$.

For the simple global kink constraint used in the reactor-design model, require $q_* > q_K$, where $q_K \simeq 2$. Solving Eq. (40) for I_p gives a current ceiling,

$$I_p < \frac{2\pi a^2 B_0}{\mu_0 R_0 q_K} \frac{1 + \kappa^2}{2}. \quad (41)$$

The direction of the result is instructive. More toroidal field B_0 , more minor cross-section, or more elongation permits more current under this particular limit; larger major radius R_0 lowers the permitted current when the other variables are held fixed.

Primary Friedberg parameter — q_* and I_p . Equation (40) is the shaped global safety factor used by Freidberg's reactor model. Equation (41) turns its stability requirement into a quantitative current constraint. Together they couple geometry (R_0, a, κ) , field (B_0) , and the plasma current (I_p) .

Model and limits. The estimate in Eq. (40), and the $q_K \simeq 2$ limit, are the large-aspect-ratio design closure adopted in Freidberg, Mangiarotti, and Minervini (2015), Eqs. (30) and (39), open manuscript cited on page 1. A full stability boundary depends on profiles, wall response, and feedback. We use the explicit model before asking it to carry any design conclusion.

Design consequence. Elongation is valuable twice in this reduced model: it increases volume and relaxes a particular current limit. But it also created the feedback demand on the preceding page. A good parameter cannot be optimized one equation at a time.

16. A single safety factor is not enough: the current profile matters

Equation (39) contains more information than the edge number q_* . If two plasmas have the same total I_p but different functions $j_\varphi(r)$, then $I_{\text{enc}}(r)$ and therefore $q(r)$ differ throughout the interior. The machine cannot treat its current as a single knob when the plasma's magnetic geometry depends on where that current flows.

The reason is already visible in the field-line picture. When $q = m/n$ for integers m and n , a field line closes after n poloidal and m toroidal circuits in the idealized description. Such rational surfaces are not automatically harmful, but perturbations can accumulate coherently there. That is why later stability discussions specify a profile such as $q(0)$ and $q(a)$, rather than only a total current.

There is also a practical lesson. The transformer acts on current induction, while auxiliary systems can add or reshape current through other mechanisms. The sources of current and the resulting profile must both be accounted for in any steady-state claim. At this stage we need not choose a detailed profile. We need to preserve the variable that the later stability and bootstrap-current arguments will require.

Here bootstrap current means a pressure-gradient-driven share of the toroidal plasma current. Its orbit-and-collision mechanism is derived in Sections 36–37, before any numerical bootstrap fraction is used.

Supporting quantity — $q(r)$ and $j_\varphi(r)$. The radial safety-factor profile in Eq. (39) carries the current-distribution information that a single q_* loses. Later closure arguments will need it to distinguish total current from the fraction supplied by the plasma itself or by external drive.

FIGURE GUIDE

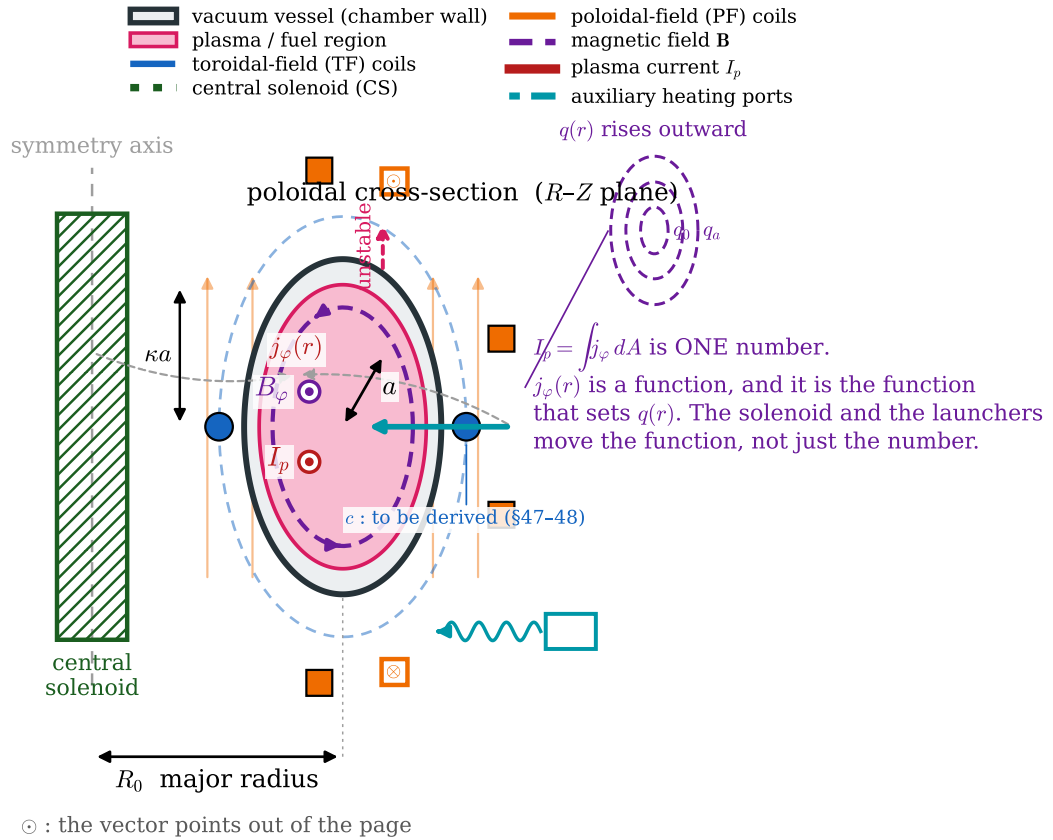
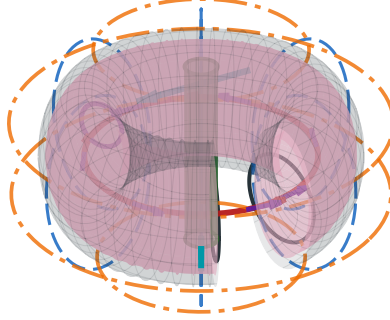
The current-profile construction diagram appears on the following page.

Design consequence. The next design calculations must carry both a global current I_p and a profile $q(r)$. That added bookkeeping is not mathematical fussiness: it records which field lines and which stability limits the plasma actually contains.

Stage 7a — total current is not the same as its profile

the solenoid and the launchers do not just set I_p ; they set where the current sits, $j_\varphi(r)$, and that is what fixes $q(r)$

perspective view (vessel cut open)



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 8: Machine state: the current profile. Total current and its distribution $j_\varphi(r)$ are different quantities, and it is the profile that fixes $q(r)$. No new hardware appears — this is an annotation layer.

17. Why field strength and volume dominate the first performance estimate

A fusion reaction needs one deuterium ion and one tritium ion. In a roughly equal D–T mixture, the number of reaction opportunities per unit volume is proportional to n^2 , where n is a representative fuel density. At a fixed useful temperature, the reactivity $\langle\sigma v\rangle$ is also fixed, so

$$P_{\text{fus}} \propto n^2 V. \quad (42)$$

This is a scaling statement, not yet a reactor-power calculation: the missing proportionality constant contains the reaction energy and $\langle\sigma v\rangle(T)$.

For a fully ionized, equal-temperature D–T plasma, ions and electrons both contribute to pressure, so $p \simeq 2nT$. The beta definition from [Sec. 8](#) gives $p = \beta B_0^2 / (2\mu_0)$ in this reduced scaling model. Therefore, at fixed T ,

$$n \propto \beta B_0^2, \quad P_{\text{fus}} \propto \beta^2 B_0^4 V \propto \beta^2 B_0^4 R_0 a^2 \kappa. \quad (43)$$

The fourth power of field is the memorable point. If all the stated assumptions hold, raising B_0 slightly has a much larger effect on this estimate than raising it by the same percentage appears to have at first glance. That is a motivation, not permission, to choose an arbitrarily large field: coils, stresses, superconductors, plasma limits, and the central stack will later impose the corresponding prices.

Primary Friedberg parameter — B_0 and β . Equation (43) is the first bridge from plasma equilibrium to reactor output. It explains why the toroidal field B_0 , normalized pressure β , and shaped volume $2\pi^2 R_0 a^2 \kappa$ recur throughout Friedberg’s design closure. The equation is a conditional scaling law; temperature, confinement, and stability have not been eliminated.

Model and limits. This page holds T , fuel mix, and $\langle\sigma v\rangle$ fixed, identifies the confining field with B_0 , and ignores profile factors. The actual D–T reactivity varies strongly with temperature. The purpose is to derive the dependence that remains once a target operating temperature has been chosen, not to replace a reactivity model.

FIGURE GUIDE

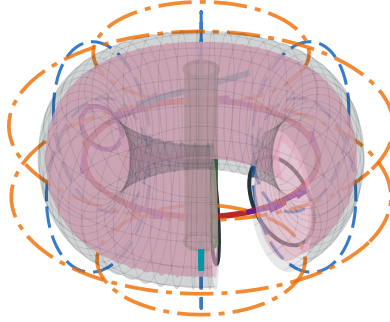
The performance-quantity diagram appears on the following page.

Design consequence. The machine now has a reason to seek strong field, high allowed pressure, and useful volume. The remainder of the thesis must show whether these desires can coexist with current limits, confinement, materials, coil technology, and power balance.

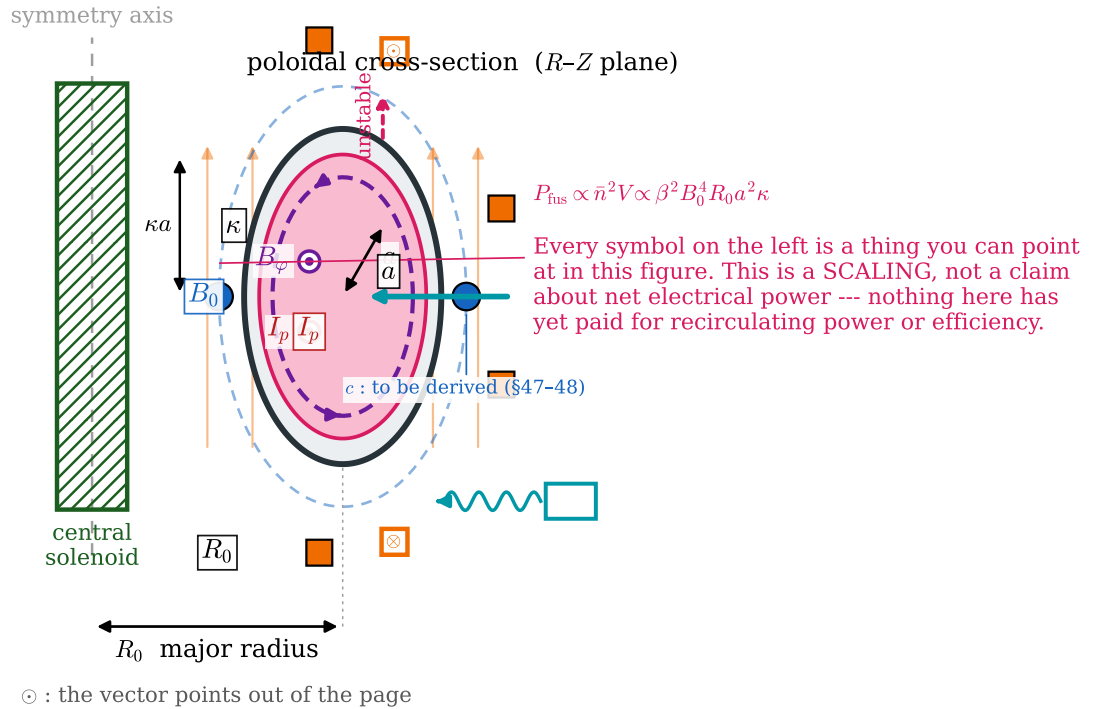
Stage 7b — the performance quantities live on the machine

every symbol in the fusion-power scaling is a feature you can point at

perspective view (vessel cut open)



- | | |
|--|---|
|  vacuum vessel (chamber wall) |  poloidal-field (PF) coils |
|  plasma / fuel region |  magnetic field $\mathbf{B}$ |
|  toroidal-field (TF) coils |  plasma current I_p |
|  central solenoid (CS) |  auxiliary heating ports |



Idealized schematic; not a fabrication drawing.

$R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 9: Machine state: the performance quantities, attached to the machine. Every symbol in the fusion-power scaling is a feature that can be pointed at.

18. A ledger before optimization: what has actually been derived?

These pages have built a ledger of demands rather than selected a tokamak by declaration. It prevents improving one formula while silently violating another.

Quantity	What it measures	Why it cannot be chosen alone
$R_0, a, A = R_0/a$	Overall and cross-sectional size	Set geometry, volume, and aspect ratio.
B_0	Toroidal confining field	Shrinks gyro-orbits; appears as B_0^4 in the conditional power scaling.
I_p	Toroidal plasma current	Creates poloidal field, but is bounded by the safety-factor constraint.
$q(r), q_*$	Field-line pitch and its shaped global measure	Depend on current distribution as well as total current.
κ	Vertical elongation	Increases volume and the reduced I_p allowance; requires shaping and feedback.
n, T, p, β	Fuel state and magnetic-pressure fraction	Set reaction opportunity and challenge confinement and stability.
τ_E	Energy-confinement time	Decides whether heating can outrun losses; it remains to be derived and modeled.

The drawing now has earned components: vacuum and magnetic field for hot fuel; a torus to remove open ends; a solenoid and plasma current for the poloidal field; PF coils for position and shape; feedback for vertical stability; and heating/current-drive ports because the transformer has a finite clock.

FIGURE GUIDE

The first construction-atlas checkpoint appears on the following page.

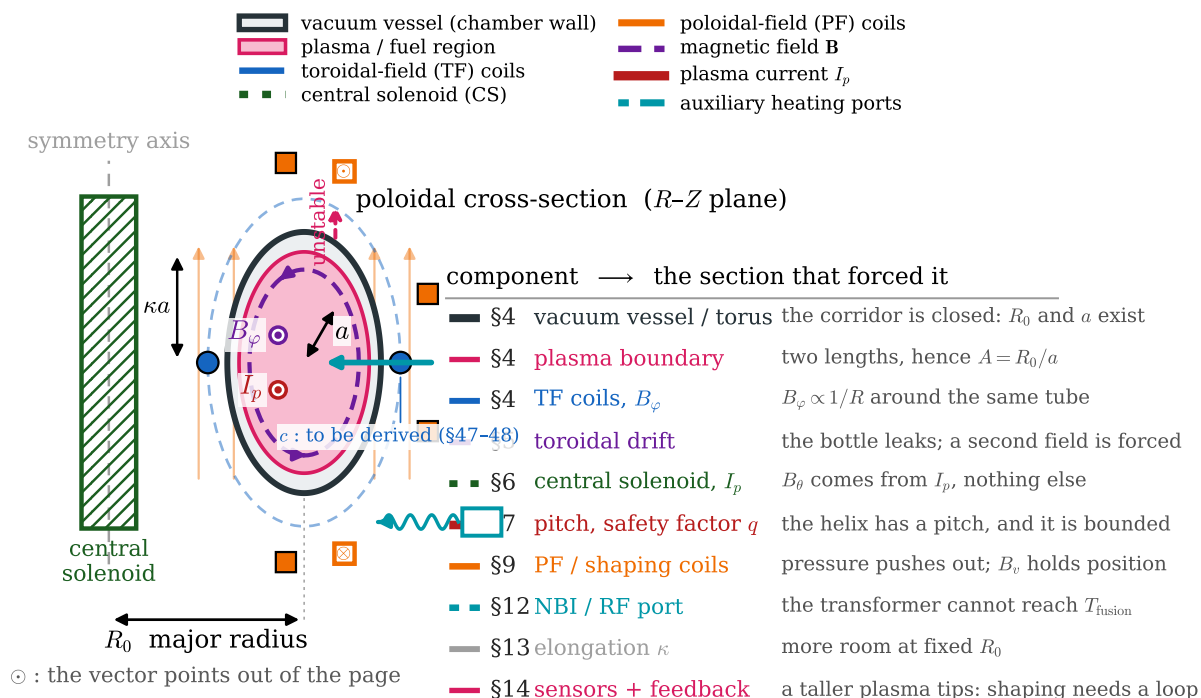
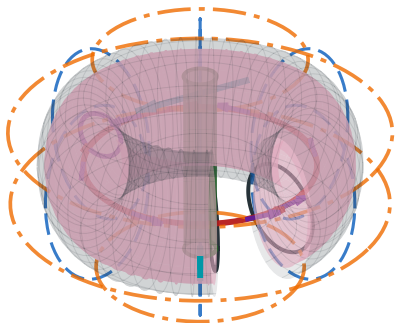
Supporting quantity — first coupled ledger. The variables in this ledger are the minimum state carried forward into the Friedberg-style closure problem. A proposed parameter set is acceptable only when the equations that constrain *all* of them are checked together.

Design consequence. The ordinary tokamak is now visible, but only as the accumulated answer to specific physical failures. The next section can add the remaining reactor subsystems in the same discipline: name the failure, derive the demand, then add the component and its variables to the ledger.

Atlas checkpoint — what has actually been derived so far

every component on the machine at this point, numbered by the section that forced it. Nothing later appears.

perspective view (vessel cut open)



NOT here, because not yet derived: triangularity, alpha heating, the neutron boundary, blanket, shield, superconductors, divertor, and every spherical-tokamak feature. A component enters this table only after a section has forced it.

Idealized schematic; not a fabrication drawing.
 $R_0/a = 3.0$, $\kappa = 1.70$, $\delta = 0.00$, 6 of 12 TF coils in the 3-D panel; all geometry computed by tokamak_build.py.

Figure 10: Atlas checkpoint. Every component derived so far, numbered by the section that forced it. Nothing later than §18 appears: a part enters this table only after a section has made it necessary.